



Information Technology
Generalist

PROG1400 – Introduction to Object-Oriented Programming

Course Summary

PROG1400 introduces students to object-oriented programming (OOP) using a structured **project-based learning (PBL)** model. Throughout the semester, students develop a Pac-Man-style game application while the instructor models concepts using the classic Pac-Man case study. Students apply each newly introduced concept—such as encapsulation, abstraction, inheritance, and polymorphism—directly to their own evolving project.

The course integrates:

- **UML modelling** (class and sequence diagrams)
- **Python OOP implementation**
- **Version control using GitHub**
- Iterative design refinement from concept to final product

Pedagogical Importance

This course emphasizes learning through **applied practice and iterative development**, rather than isolated exercises. Students build one evolving project across the semester, reinforcing conceptual understanding through repeated and phased implementation. The structured progression from modelling → coding → refinement strengthens analytical thinking, problem-solving, and design literacy.

The midterm practical exam is a hands-on practical evaluation assessing modelling, implementation, and professional workflow skills, ensuring students demonstrate applied competency rather than memorization.

There is no final exam as the student will spend the remainder of the term, implementing, modelling and refactoring (continuous improvement) their game of choice re-enforcing the concepts and use of tools introduced in the first part of the course.

For the final evaluation students will be required to reflect upon their progression, identify the challenges and be able to describe the solutions to these challenges in the context of OOP. In addition, the final evaluation will measure the application of OOP principles throughout their game of choice.

Equitable and Inclusive Approach

The course is designed to support diverse learning styles by combining:

- Visual modelling (UML diagrams)

- Guided case-study instruction (Pac-Man demonstrations)
- Incremental scaffolding and milestone feedback
- Frequent low-stakes assessments before major evaluations
- Portfolio-based project development

Students select their own game concept, allowing personal interest and creativity to drive engagement while maintaining consistent academic standards. Continuous feedback and structured checkpoints promote student success and reduce high-stakes pressure.

This course prepares students for advanced programming studies and collaborative software development environments while fostering inclusive, applied, and student-centered learning.